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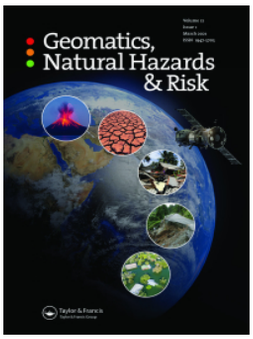
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Spatiotemporal economic impact analysis of the Taal Volcano eruption using electricity consumption and nighttime light data

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ABSTRACT

Analyzing the spatiotemporal dimension of the economic impacts of disasters is critical for providing timely and proportionate support. However, traditional economic impact measures often lack spatiotemporal details. Hence, we estimated the daily electricity consumption (EC) loss using high-frequency EC data and analyzed temporal dimensions of the immediate impacts of Taal Volcano's eruption on January 12, 2020. Subsequently, we computed the nighttime light (NTL) change using high-resolution NTL data to analyze the spatial distribution of these impacts. The temporal analysis revealed two EC loss peaks. The first peak coincided with the power outage and decreased energy demand, followed by a brief recovery after power restoration. The second peak emerged upon the influx of the affected population, evacuees, earthquake occurrences, and road closures. Major assistance distribution spurred recovery towards the end of the analysis period. The spatial analysis showed significant NTL reductions where ashfall intersected built-up areas. The municipalities with the worst NTL reduction included Talisay, Laurel, Santa Teresita, Agoncillo, Lemery, San Nicolas, Taal, and Balete. This study demonstrates an alternative method that uses high-frequency EC to reveal daily temporal fluctuations of immediate non-agricultural economic impacts and map out the areas worst affected by a volcanic eruption.

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SNPP VIIRS DNB; Philippines; regression; remote sensing; disaster loss and damage; Taal Volcano 2020 eruption

1. Introduction

Spatiotemporal dimensions are critical for the proportionate and timely distribution of resources in disaster-affected areas. Assessments of volcanic eruption impacts, however prompt, often lack these spatiotemporal dimensions. Spatiotemporally blind

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economic impact assessment may stem from a lack of data. Typical measures like GDP data are available at the national and quarterly scale, often aggregating the impacts across large areas. Further, immediate on-site data collection can be challenging due to safety reasons. Consequently, temporally blind assessments could delay a response and conceal recovery trends, while spatially blind estimates lead to possible mismatched and biased resource allocation (Brucal et al. 2020; Malik 2022).

A possible solution to this lack of data is to use alternative high-frequency electricity consumption (EC) and high-resolution nighttime light (NTL) datasets. Both datasets exhibit a strong relationship with GDP, as shown in the regression model results (Online Appendix A) and previous studies (Doll et al. 2000; Ghosh et al. 2010; Henderson et al. 2012; Al-mulali and Mohammed 2015; Faisal et al. 2017; Khan and Kong 2020; Chen et al. 2022).

Several studies have used EC data to assess economic impacts of the COVID-19 pandemic (Fezzi and Fanghella 2020; Beyer et al. 2021; Wang et al. 2021; Ai et al. 2022; Del Castillo et al. 2024). EC data also showed a decline following a volcanic eruption (Wilson et al. 2012). However, no research has yet applied this data to estimate the impacts of local disasters.

NTL data have been used to map economic impacts of local disasters such as typhoons, earthquakes, and floods (Qiang et al. 2020; Tveit et al. 2022; Jia et al. 2023; Zhou et al. 2024). Although NTL data's continuous acquisition is ideal for long-term impact analysis of volcanic eruptions, no studies have yet utilized NTL to map the impacts of volcanic eruptions.

This study uses EC and NTL data to analyze spatiotemporal distributions of the economic impacts of the Taal Volcano eruption on January 12, 2020. It adopts the following research questions: (1) How were the economic impacts based on EC data distributed across time, and (2) which municipalities were worst affected, based on NTL data?

To answer the first research question, we estimated the daily economic impacts using daily EC data. First, we generated a regression model of pre-eruption daily EC to estimate a counterfactual EC, then subtracted the actual EC from it to compute the daily EC loss (ECL) as a measure of economic impact. Assuming that actual EC data were only affected by the eruption, the analysis covered 18 days, ending on January 30, 2020, when COVID-19 was first detected in the Philippines. Extending the period could introduce pandemic-related influences beyond the study's scope. For the second research question, NTL was used to identify the most affected municipalities by computing the NTL change (NTLC) as the difference between the average NTL of previous years and the eruption month's NTL.

This study presents a novel approach to estimating the immediate impacts of volcanic eruptions. To our knowledge, no study has used EC data to assess the economic effects of natural disasters, making this the first to apply EC data for short-term impact analysis after a local disaster. Additionally, our analysis using NTL data identifies the most affected municipalities and is the first to use NTL data to map the impacts of a volcanic eruption. While EC and NTL data were analyzed separately due to resolution differences, we propose a method for remotely assessing the daily and municipal distribution of economic impacts, which can be applied even while significant hazards persist onsite.

The remainder of this article is organized as follows. Section 2 offers background information on the Taal Volcano, and Section 3 describes the datasets and methods used. Sections 4 and 5 present and discuss the results, respectively. Section 6 concludes the article.

2. The 2020 eruption of the Taal Volcano

The Taal Volcano, located in the Southern Luzon Region of the Philippines (Figure 1), erupted phreatomagmatically in 2020, affecting an estimated 8,600 km^2 with ashfall (Balangue-Tarriela et al. 2022) and impacting over 700,000 people (NDRRMC 2020). The eruption caused significant damage to agriculture, aquaculture, residences, and infrastructure, disrupted transportation, education, work, air travel, and affected the air quality (Del Castillo et al. 2020; Lim et al. 2022; Santos et al. 2023). It resulted in damage costing over 8.4 billion PhP (NEDA IVA, 2020). Weaker explosions continued for 10 days (Balangue-Tarriela et al. 2022). The most affected areas included Agoncillo, Alitagtag, Balete, Cuenca, Laurel, Lemery, Lipa City, Malvar, Mataas na Kahoy, San Nicolas, Santa Teresita, Taal, Talisay, Tanauan, and Tagaytay (NEDA IVA, 2020).

3. Materials and methods

3.1. Materials

3.1.1. EC

EC represents the total energy used over a given period. For this study, the daily EC covers the hourly electric load data from the National Grid Corporation of the Philippines (NGCP, 2023), which tracks EC across three grids: Luzon, Visayas, and Mindanao (Figure 2). Electricity loads per hour h and grid g $EL_{g,h}$ were summed to obtain total EC per grid $EC_{g,d}$ per day d (Equation 1).

$$EC_{g,d} = \sum_{h=1}^{24} EL_{g,h} \tag{1}$$

We focused on the Luzon electric grid which covers the Taal Volcano. This grid includes both affected and unaffected regions. Due to the lack of daily EC data with finer spatial resolution, we assumed that EC in unaffected regions remained normal. Any reduction in EC was attributed solely to the eruption, as it was the only significant event for that month on the island during the study period.

Following Equation (1), daily Luzon EC was computed from January 1, 2013 through January 30, 2020. Figure 3 shows an ascending daily EC trend and noticeable irregularities that could indicate power interruptions, disaster events, or holiday periods such as Christmas, New Year, and other annual religious and cultural occasions.

Figure 4 plots the average daily EC in Luzon across months in 2019. As shown, EC peaks in the warmest months of May and June and plummets in the dry cold months of January and February. Generally, it fluctuates weekly, peaking on weekdays or workdays and dropping on rest days (i.e. Sundays).

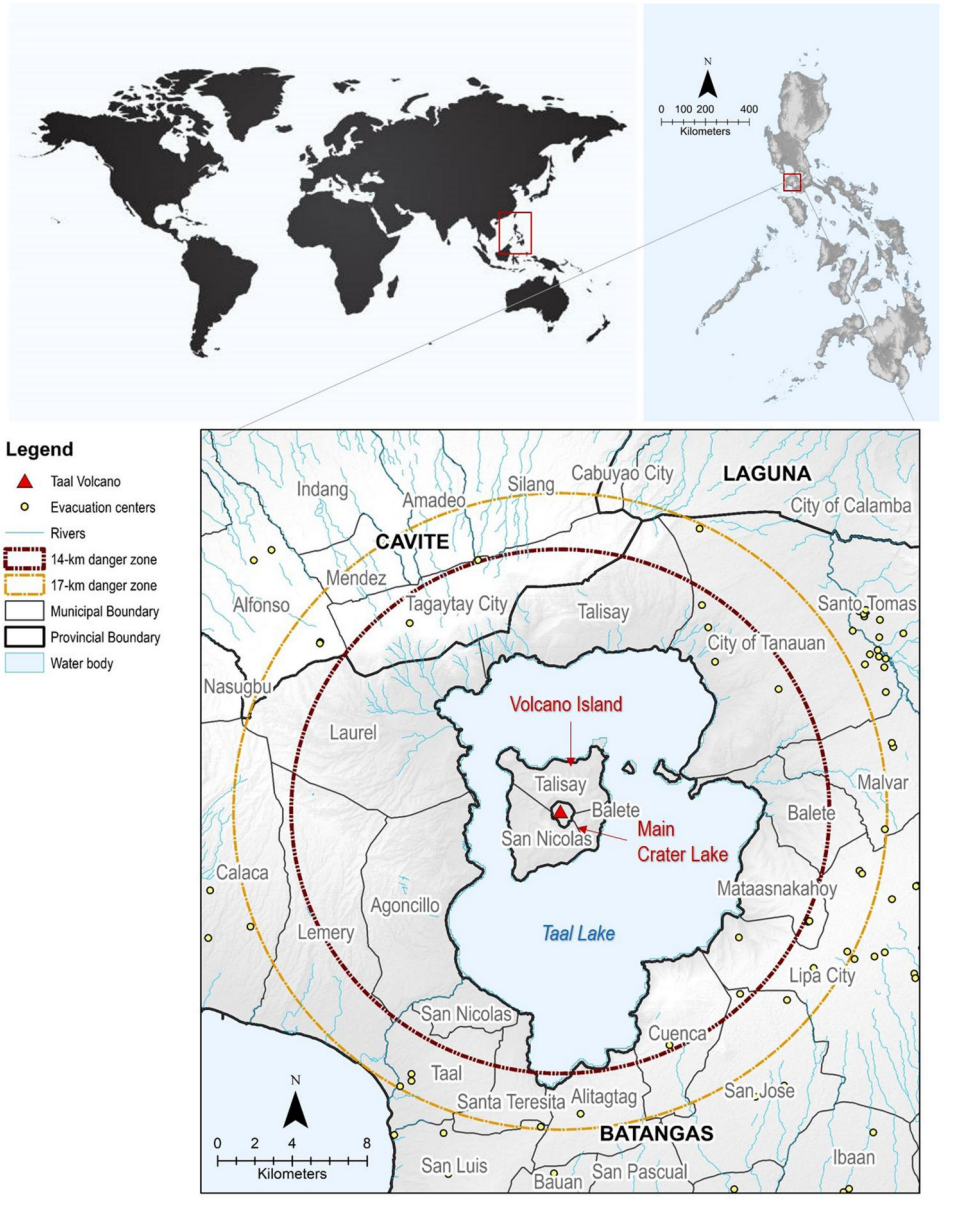


Figure 1. Location of Taal Volcano and municipalities within the 17-km danger zone.

3.1.2. Temperature

The temperature is a significant variable in residential energy consumption in the Philippines (Bayudan-Dacuycuy 2017; Dacuycuy 2019; Santos 2020; Sabado et al. 2022). Up to 56% of household energy consumption in the country comes from space cooling and refrigeration (Department of Energy 2018), both related to temperature.

We collected the daily mean temperature data per monitoring station from the Philippine Atmospheric, Geophysical, and Astronomical Services Administration (PAGASA 2023). The average of the daily mean temperature of all the monitoring

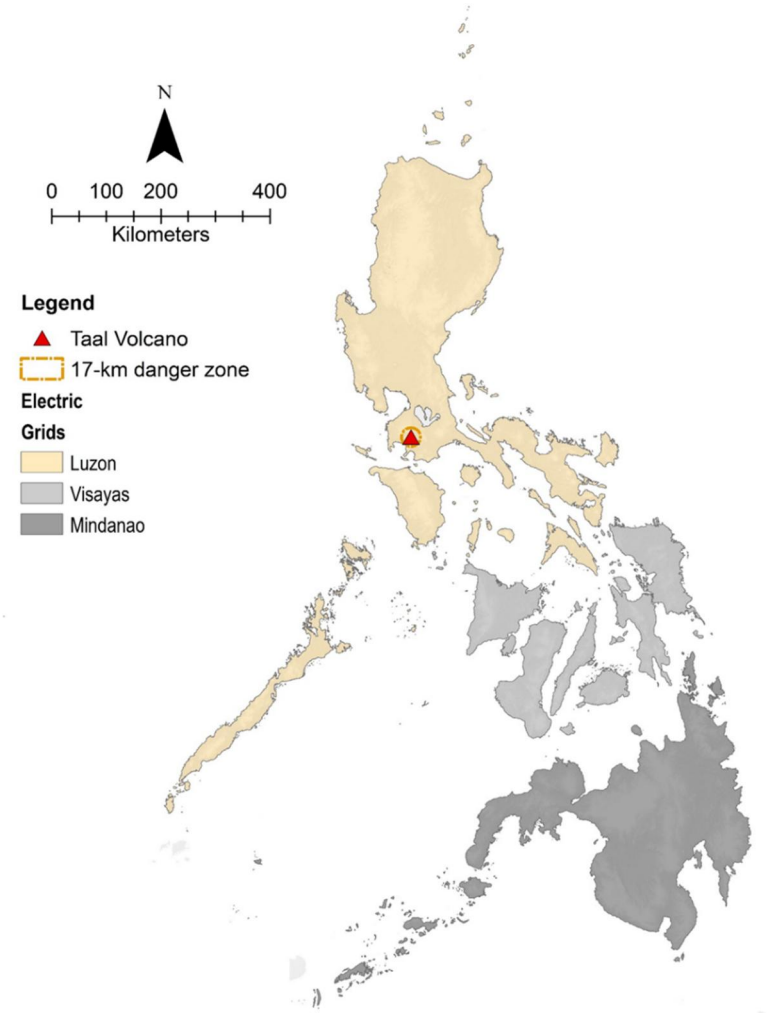


Figure 2. Electricity grids in the Philippines and Taal Volcano location.

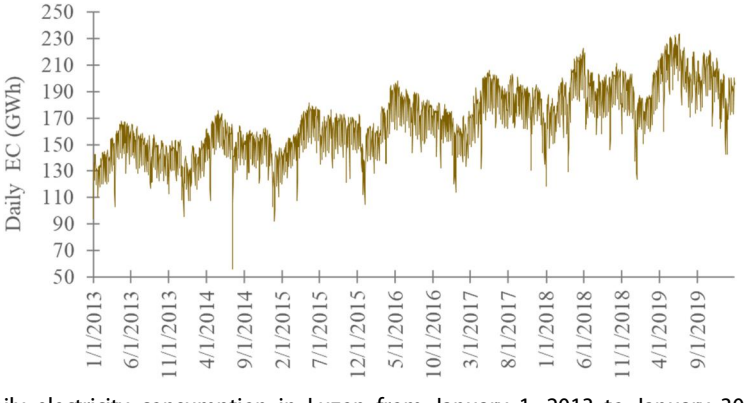


Figure 3. Daily electricity consumption in Luzon from January 1, 2013 to January 30, 2020. Data source: (NGCP 2023).

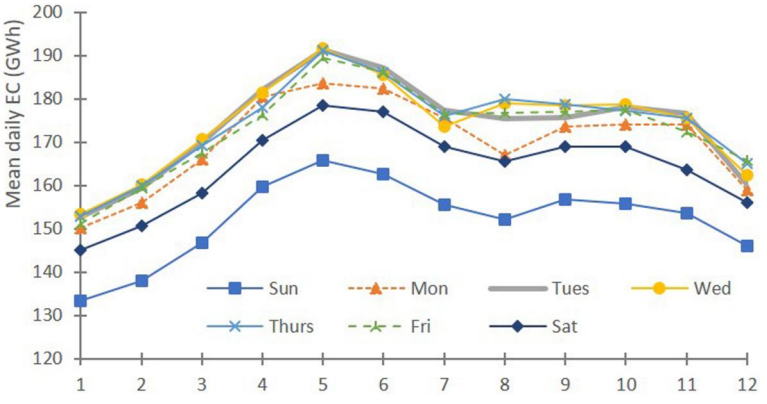


Figure 4. Daily average EC in Luzon per month from January 1, 2019 to December 31, 2019. Data source: (NGCP 2023).

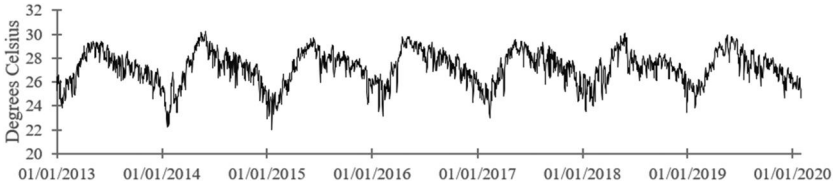


Figure 5. Daily mean temperature in Luzon. Data coverage: January 1, 2013 to January 30, 2020. Data source: (PAGASA 2023).

stations in Luzon $meantemp_{L,d}$ was computed (Equation 2) to match the scale of EC data.

$$meantemp_{L,d} = \frac{\sum_{s=1}^{n} temp_{s,d}}{S} \tag{2}$$

where, $temp_{s,d}$ is the mean temperature of monitoring station s on day d over the total number of monitoring stations in Luzon S.

Figure 5 presents the $meantemp_{L,d}$ data from January 1, 2013, to January 30, 2020, showing how the mean temperature peaks from April to June and drops in January and February, coinciding with the daily EC patterns.

3.1.3. NTL

NTL is a three-decade-long satellite dataset of global nocturnal light emissions from natural and artificial sources. NTL is valuable as a proxy for mapping economic activity (Ghosh et al. 2010) and estimating disaster impacts (Tveit et al. 2022) and recovery (Qiang et al. 2020) because it reveals the spatial details of an economy (Zheng et al. 2023) due to its ~500 m resolution. However, its application, which is still limited in the Philippines, has focused on determining the impacts of typhoons and floods (Strobl 2019; Cheng and Han 2022; Pagaduan 2022).

For this study, we downloaded NTL data captured by the Suomi National Polar-Orbiting Partnership Visible Infrared Imaging Radiometer Suite Day and Night Band

sensor and pre-processed by the Earth Observation Group of the Payne Institute for Public Policy, Colorado School of Mines (Elvidge et al. 2013; Earth Observation Group 2021). Pre-processing removed the extraneous luminosity (Elvidge et al. 2017) and composited the images into monthly and annual datasets (Elvidge et al. 2021). Following Del Castillo et al. (2024), these were further processed to remove negative values and address the noise and cloud issues *via* filtering methods. We used monthly, instead of daily data, to reduce the effects of clouds or ashfall obstructing light detection after the eruption.

Figure 6 shows the pre-processed NTL data from January 2017 to January 2020. The blue areas indicate low NTL values, while yellow areas indicate high NTL values. Comparing these annual images reveals changes in nighttime brightness. Post-eruption dimming occurred within the 14-km danger zone, particularly in Talisay, Laurel, Agoncillo, Taal, and San Nicolas. Conversely, bright areas expanded outside the 17-km danger zone, especially in Tanauan, Calamba, Malvar, Mataas na Kahoy, and Lipa City.

3.2. Methods

We used high-frequency EC and high-resolution NTL datasets as proxies for economic activity to analyze the spatiotemporal distribution of the eruption impacts. Initially, we examined the relationship of EC and NTL with a usual measure of economic activity, being GDP. Due to the lack of high-frequency and high-resolution GDP data, we aggregated the EC and NTL data to the national quarterly scale and performed a regression analysis (Online Appendix A). The results indicated that EC and NTL effectively explain national quarterly GDP. Additionally, we found a strong relationship between EC and NTL, reinforcing their use as measures of economic activity.

Based on the established relationship, we analyzed the temporal and spatial distribution of Taal Volcano's economic impacts using daily EC and monthly NTL datasets. Due to differing data scales, we conducted separate analyses to maximize spatial and temporal details from each data. We analyzed the temporal distribution of impacts using daily EC data through electricity consumption loss (ECL) estimation involving three steps: pre-eruption EC modeling, counterfactual EC estimation, and ECL computation. Subsequently, we analyzed the spatial distribution of impacts using monthly NTL data through NTLC computation, aggregation, and mapping.

3.2.1. ECL estimation process

Figure 7 illustrates the ECL estimation process. We estimated the daily economic activity loss in terms of ECL by subtracting the actual EC from the counterfactual EC estimated using a model generated based on the pre-eruption EC data. This model established the relationship between the explanatory variables and pre-eruption EC under normal conditions. This relationship would persist had there been no eruption. Under this condition, we use the pre-eruption model to estimate the counterfactual EC. Then, ECL was computed by subtracting the actual from the counterfactual.

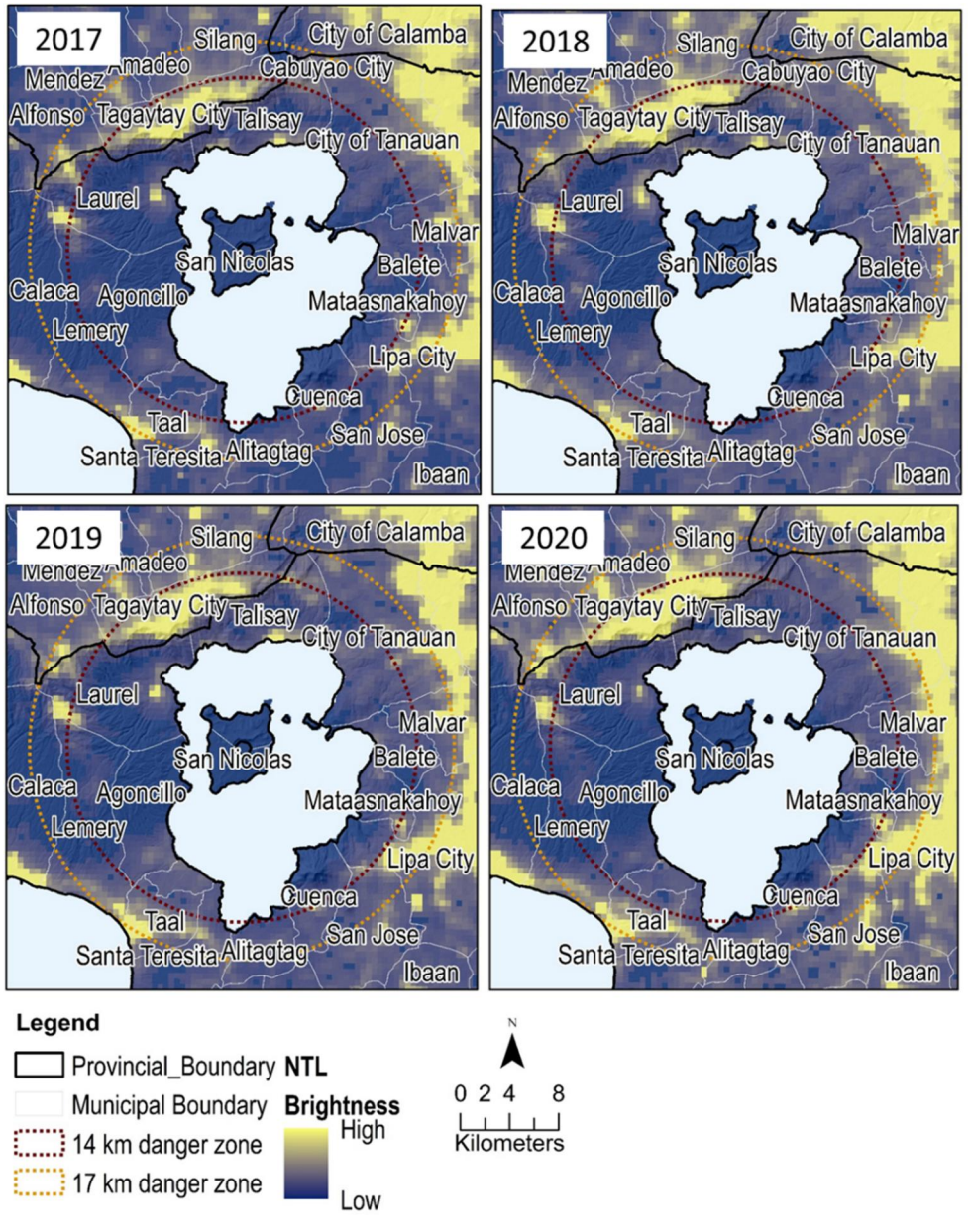


Figure 6. Pre-processed NTL data for January 2017 to January 2020.

This approach resembles the method of Beyer et al. (2021), who assessed the impact of COVID-19 lockdowns in India. They demonstrated that national quarterly EC and NTL data could serve as proxies for GDP. To estimate daily pre-pandemic EC by district, they used maximum temperatures for cooling and heating and incorporated quarter-fixed effects and trend components to adjust for seasonal variations and time effects. Using this model, they predicted daily EC during the pandemic and compared it to actual EC to measure the impacts.

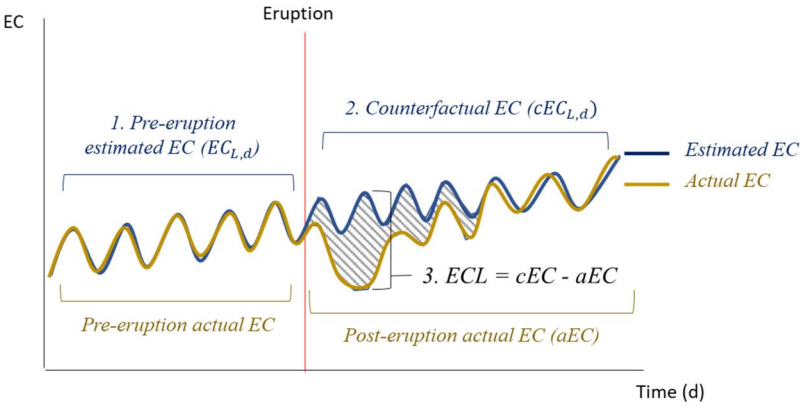


Figure 7. Illustration of the ECL estimation process.

We used mean temperature instead of maximum temperature for cooling and heating due to the positive linear relationship between EC and temperature, unlike the U-shaped correlation observed in India (Online Appendix B). Key features of this study include validating the method from previous research and extending its application to a different study site and to the context of a volcanic eruption.

3.2.1.1 Pre-eruption EC modeling. The first step in the ECL estimation process was to generate the pre-eruption EC model that captured the relationship between the dependent variable EC and the independent variables before the eruption. For this, we divided the daily Luzon datasets into pre- and post-eruption subsets.

We regressed the natural logarithm of the pre-eruption EC $lnEC_{L,d}$ for Luzon L per day d on the natural logarithm of mean temperature $lnmeantemp_{L,d}$ using pre-eruption data. Mean temperature was chosen as an explanatory variable due to its influence on energy use (Bayudan-Dacuycuy 2017; Dacuycuy 2019; Santos 2020; Sabado et al. 2022) and its stability during the eruption (Ramirez 2020), enabling a counterfactual or "business-as-usual" estimate. EC patterns varied by day (Figure 4), with higher consumption on weekdays and lower consumption on weekends. To account for this effect, we included the day-of-the-week dummy variable *DW*. We also included a holiday dummy variable *Holiday* to capture the reduced EC on national holidays (Figure 3), coding holidays as 1 and non-holidays as 0. Certain holidays and seasons, such as the Holy Week and Christmas, have extended effects, which were accounted for by adding a week of the year fixed effects *WYfe*. Additionally, we included a *trend* variable to account for the annual increase in EC, as shown in Figure 3. The resulting pre-eruption model was then represented in Equation (3), where the parameters to be estimated were denoted by β_0 and β_1, and ε was the error term.

$$lnEC_{L,d} = \beta_0 + \beta_1 lnmeantemp_{L,d} + Holiday + DW + WYfe + trend + \varepsilon \quad (3)$$

3.2.1.2 Counterfactual EC estimation. In the second step, we estimated the counterfactual EC, or *cEC*, which was the EC if the eruption had not occurred. Without the eruption, we assumed that the relationship between EC and its explanatory variables

in Equation (3) would remain the same. Thus, we plugged the post-eruption data into the model represented by Equation (4) to compute the *cEC*.

$$lncEC_{L,d} = \hat{\boldsymbol{\beta}}_0 + \hat{\boldsymbol{\beta}}_1 lnmeantemp_{L,d} + Holiday + DW + WYfe + trend \tag{4}$$

We estimated the natural logarithm of counterfactual EC $lncEC_{L,d}$ based on the natural logarithm of the post-eruption's mean temperature. $\hat{\boldsymbol{\beta}}_0$ and $\hat{\boldsymbol{\beta}}_1$ were the coefficients estimated using Equation (3).

3.2.1.3 ECL computation. The last step was to estimate the $ECL_{L,d}$, by taking the difference between the actual EC $aEC_{L,d}$ and the counterfactual EC $cEC_{L,d}$, computed from $lncEC_{L,d}$. ECL measures the economic impact per day through EC as an indicator of economic activity based on its relationship with GDP. Equation (5) provides the economic impact in terms of ECL per day d in Luzon.

$$ECL_{L,d} = cEC_{L,d} - aEC_{L,d} \tag{5}$$

3.2.2. NTLC analysis

NTLC analysis determines the spatial distribution of economic impacts using monthly NTL data as an indicator of economic activity based on its relationship with GDP. NTLC measures the change in nocturnal brightness from a pre- to post-eruption time.

3.2.2.1 NTLC computation. The existing literature offers two main approaches in detecting changes: difference and ratio. In the difference approach, NTL in the disaster month is subtracted from either the previous year's NTL, the previous month's NTL (Lin and Rybnikova 2023), or the average NTL of prior years (Beyer et al. 2021). The ratio approach calculates the post-disaster NTL as a portion of the pre-disaster NTL (Sarkar 2021).

We carried out all these methods mentioned above but found that the difference between the eruption month and the average of the previous years had the most meaningful results. Therefore, we computed $NTLC_p$, which represented the change in emitted light brightness per pixel p by subtracting the average NTL per pixel for January 2017 $NTL_{p,2017}$, January 2018 $NTL_{p,2018}$, and January 2019 $NTL_{p,2019}$ from the NTL in January 2020 $NTL_{p,2020}$ (Equation 6).

$$NTLC_p = NTL_{p,2020} - \left(\frac{NTL_{p,2017} + NTL_{p,2018} + NTL_{p,2019}}{3}\right) \tag{6}$$

3.2.2.2 Municipal NTLC aggregation. $NTLC_m$ was computed from the sum of the NTLC values of all the pixels p in municipality m (Equation 7) using zonal statistics. The resulting value allowed for the comparison of economic impacts across municipal units to determine the municipality with the greatest NTL reduction.

$$NTLC_m = \sum_{p=1}^{n} NTLC_p \tag{7}$$

4. Results

4.1. Pre-eruption EC model

Four pre-eruption EC regression models were generated to compare the explanatory power of the variables (Table 1). The lnEC1 model showed that the *trend* could explain 69.7% of the pre-eruption EC. Adding the *DW* dummy variables in lnEC2 increased the R^2 to 0.808. Including the *Holiday* dummy variable in lnEC3 further raised the R^2 to 0.859. The addition of the *lnmeantemp* variable in lnEC4 boosted the explanatory power of the model, explaining 90.5% of the pre-eruption EC. Of the four models, lnEC4 had the highest R^2 value and was used to estimate the counterfactual EC.

4.2. Counterfactual EC

Post-eruption data were plugged into the lnEC4 model to estimate cEC. The computed cEC is for the entire Luzon Island. Table 2 reports the daily cEC estimates.

The computed cEC was plotted with aEC for comparison. Figure 8 shows that cEC is higher than aEC in the first two weeks after the eruption, indicating loss. However, aEC increased in the third week exceeding the cEC, suggesting recovery.

4.3. Daily ECL

Subtracting aEC from cEC yielded the daily ECL. ECL was highest on the day after the eruption (Figure 9), briefly recovered on days 6 and 7, increased again on day 8 reached its second peak on day 11, dropped on day 12, and further dropped to negative values on day 13, only to rise again the following day. Subsequently, ECL decreased until the end of the observation period.

Table 1. Regression models for pre-eruption EC estimation.

	lnEC1	lnEC2	lnEC3	lnEC4
trend	0.057***	0.057***	0.058***	0.057***
	(0.001)	(0.001)	(0.001)	(0.0004)
MON		0.112***	0.117***	0.116***
		(0.005)	(0.004)	(0.003)
TUE		0.133***	0.133***	0.133***
		(0.005)	(0.004)	(0.003)
WED		0.134***	0.134***	0.133***
		(0.005)	(0.004)	(0.003)
THURS		0.135***	0.136***	0.135***
		(0.005)	(0.004)	(0.003)
FRI		0.128***	0.130***	0.130***
		(0.005)	(0.004)	(0.003)
SAT		0.077***	0.079***	0.080***
		(0.005)	(0.004)	(0.003)
HOLIDAY			−0.146***	−0.140***
			(0.005)	(0.004)
lnmeantemp				0.043***
				(0.001)
WYfe	YES	YES	YES	YES
Observations	2,567	2,567	2,567	2,567
R^2	0.697	0.808	0.859	0.905
Adjusted R^2	0.691	0.803	0.855	0.903

Data coverage: January 1, 2013 to January 11, 2020. Source: (NGCP 2023; PAGASA 2023).

Table 2. Computed counterfactual daily EC and EC loss.

Date	Days from eruption	Day of the week	cEC	aEC	ECL MWh	%ECL ECL/cEC	Cumulative ECL MWh
1/12/2020	0	Sun	173,312	172,606	706	0.41	706
1/13/2020	1	Mon	191,241	177,517	13,724	7.18	14,431
1/14/2020	2	Tue	198,935	187,122	11,813	5.94	26,244
1/15/2020	3	Wed	200,305	192,796	7,509	3.75	33,753
1/16/2020	4	Thurs	204,412	197,928	6,484	3.17	40,237
1/17/2020	5	Fri	204,258	200,594	3,664	1.79	43,901
1/18/2020	6	Sat	190,530	191,766	−1,236	−0.65	42,665
1/19/2020	7	Sun	173,917	174,723	−806	−0.46	41,859
1/20/2020	8	Mon	195,260	189,502	5,758	2.95	47,616
1/21/2020	9	Tues	197,074	191,250	5,824	2.96	53,440
1/22/2020	10	Wed	201,833	191,473	10,360	5.13	63,800
1/23/2020	11	Thurs	205,055	194,000	11,055	5.39	74,854
1/24/2020	12	Fri	197,619	190,571	7,048	3.57	81,903
1/25/2020	13	Sat	164,323	172,847	−8,524	−5.19	73,379
1/26/2020	14	Sun	178,724	173,878	4,846	2.71	78,225
1/27/2020	15	Mon	197,918	198,460	−542	−0.27	77,683
1/28/2020	16	Tues	197,165	200,771	−3,606	−1.83	74,076
1/29/2020	17	Wed	195,036	198,899	−3,863	−1.98	70,213
1/30/2020	18	Thurs	187,482	195,495	−8,013	−4.27	62,201



Figure 8. Counterfactual vs. actual EC per day.

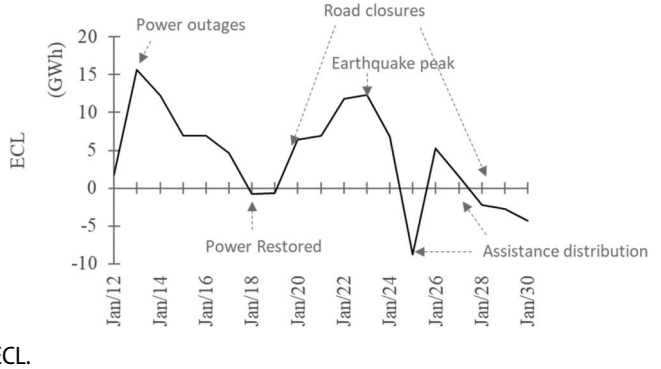


Figure 9. Daily ECL.

4.4. NTLC

We plotted the differences between pre- and post-eruption NTL by pixel to identify areas of reduced economic activity. Figure 10 shows NTL reductions in blue and increases in yellow. Most of the 14-km danger zone experienced decreased brightness, with the brightness levels of Talisay and Laurel showing intense decline, and

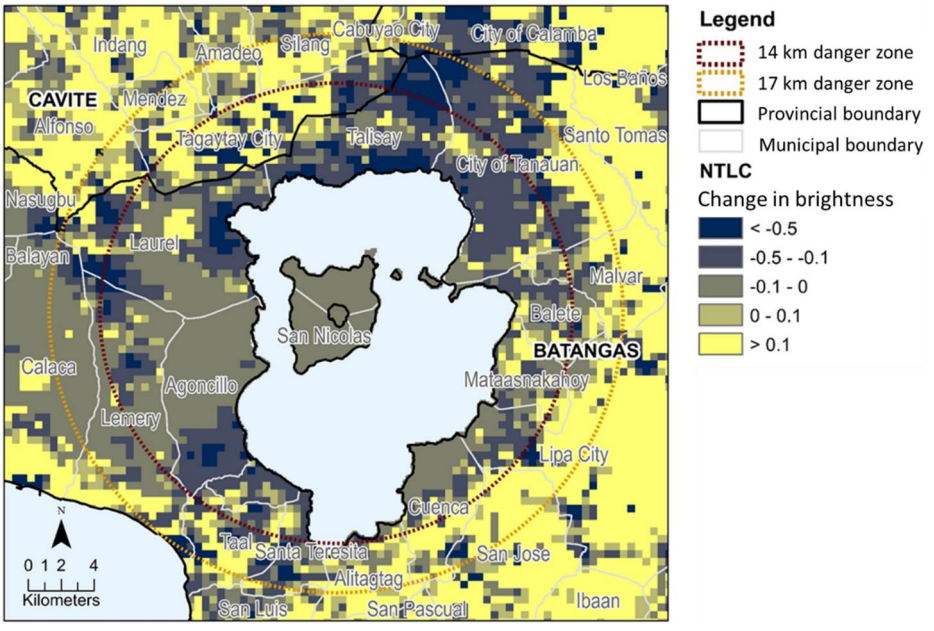


Figure 10. Computed NTLC map.

Agoncillo, Tanauan, and San Nicolas showing moderate reductions. Clusters of dimmed pixels were found between the 14-km and 17-km zones, while isolated dimmed pixels outside the 17-km zone were largely associated with decreased brightness.

We summed up the NTLC per municipality to determine the most affected units. This gave each municipality a single NTLC value. Negative NTLC sums indicated overall dimming, while positive NTLC sums suggested increased brightness. The results presented in Figure 11 show that Talisay, Laurel, and Santa Teresita had the most NTL reduction, followed by Agoncillo, Lemery, San Nicolas, Taal, and Balete. The rest of the municipalities had an overall increase in NTLC.

5. Discussion

5.1. Temporal analysis

The daily ECL's temporal pattern revealed two loss peaks and two recovery valleys. Two factors could have contributed to the first peak. One would be power disruption. Ashfall from the eruption resulted in a power shutdown, affecting 24 municipal units in Cavite, Laguna, and Batangas (NDRRMC 2020). Power outages after an eruption are common, as wet ashfall could result in substation insulation flashovers and subsequent power outages (Nagai and Nakada 2022). The second contributor to this first peak could be a decrease in demand. During the eruption, power distributors in Batangas, namely, the Batangas I and Batangas II Electric Cooperatives, reported an electricity demand reduction due to evacuation and business closures (Rivera 2020).

Power was restored in the affected municipalities after roughly six days (Santos et al. 2023) leading to a brief recovery, but ECL started rising again, reaching its

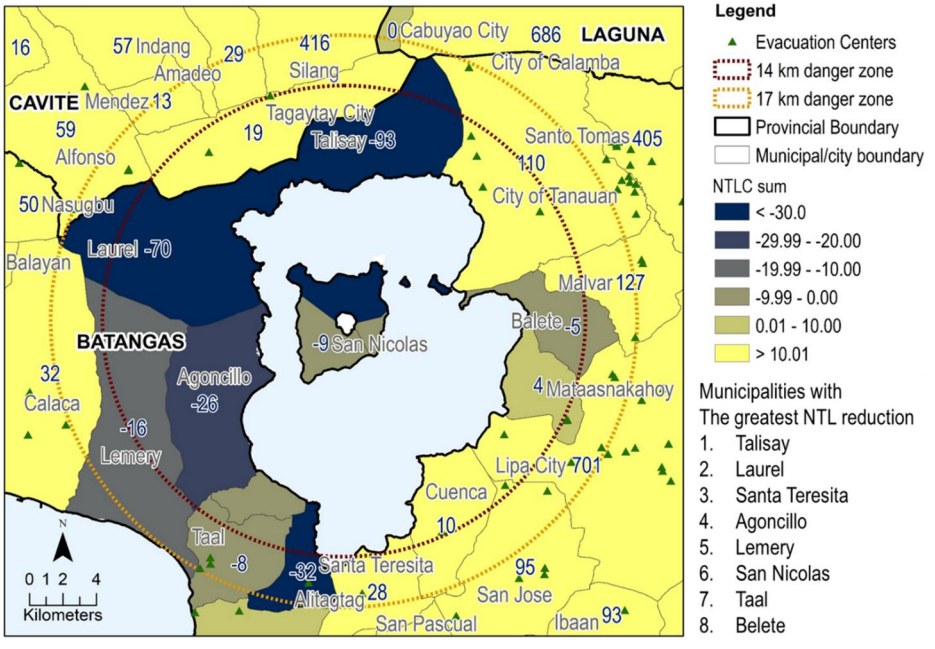


Figure 11. NTLC per municipality and location of evacuation sites. Annotated numbers show the NTLC sum.

second peak, possibly due to the increase in the affected population and evacuees (NDRRMC 2020) (Online Appendix C). Although the eruption lasted for only 10 days, earthquakes kept recurring, and up to 841 earthquakes per day were recorded on day 11 (PHIVOLCS 2020). Further, more roads were closed at the start of day 8 due to earthquake damage.

ECL dropped to its lowest level on day 13, indicating a consumption rise. This drop correlated with the distribution of financial support worth 32 million PhP that day (NDRRMC 2020), possibly stimulating increased EC in the area. The following day, ECL rose again, as assistance dropped to less than one million. On day 15, ECL dropped again, suggesting recovery, while the assistance cost increased by 22 million PhP. From this point onward, ECL continued to decline for the remainder of the study period.

Over the study period, the average %ECL (1.6%) fell between estimates from other researchers. Santos et al. (2023) estimated a pessimistic scenario through the input-output method, calculating total losses of 3.5 billion PhP over 10 days, or 1.4% of the region's annual gross regional product (GRP). A government report (NEDA IVA 2020) estimated total losses at 8.4 billion PhP, or 3.3% of the Gross Regional Product. Note that our results exclude agricultural losses, as this sector has minimal energy requirements.

Our results complement the government report by accounting for household losses. Monthly EC data for the Luzon grid (DOE 2020) disaggregates consumption into residential (35.4%), commercial (34.7%), and industrial (28.0%) sectors. Based on this, 0.57% of the computed ECL can be attributed to household loss. Unlike previous estimates (Santos et al. 2023), which show a smooth decline in economic loss, our

results reveal dynamic fluctuations in impacts, driven by the progression of the hazard, population response, and aid distribution. Beyond economic value and sectoral impacts, the temporal analysis sheds light on the population's reaction to the disaster and assistance, offering insights for responders and authorities to monitor and act swiftly. Additionally, this analysis confirmed that EC data could capture the immediate economic impacts of an eruption, extending previous research (Fezzi and Fanghella 2020; Beyer et al. 2021).

5.2. Spatial analysis

Spatial analysis revealed uneven NTL reduction, matching the 17-km danger zone, ashfall extent, and built-up areas. The most intensely dimmed spots (Figure 10) were within the 17-km danger zone. Within this zone, the dimming was concentrated in and around built-up areas (see land cover map in Del Castillo et al. 2020) within the ash-covered coasts north and west of Taal Lake (see ashfall extent map in Merin et al. 2021). Although agricultural losses have been reported (NEDA IVA 2020), NTLC was not able to detect these impacts.

Aggregating the pixels revealed the most affected municipalities, including Talisay, Laurel, Santa Teresita, Agoncillo, Lemery, San Nicolas, Taal, and Balete. This list largely aligns with reports, with two key discrepancies. First, the overestimated NTLC value in Santa Teresita was caused by an anomalous cluster of high NTL pixels in the January 2017 image, which inflated pre-eruption brightness and led to a significant NTL drop in 2020. Since the anomaly occurred in 2017, it was unrelated to the eruption. Second, our list missed several municipalities identified in government reports (NEDA IVA 2020), including Alitagtag, Cuenca, Lipa City, Malvar, Mataasnakahoy, Tanauan, and Tagaytay. This could be due to increased NTL emissions from evacuation sites in those areas. Despite these discrepancies, our NTLC map and the municipalities identified by our method demonstrate the potential of NTL data for mapping volcanic eruption impacts with fine spatial detail.

5.3. Policy and research recommendations

The temporal analysis implies the importance of prompt lifeline restoration and financial-support distribution for recovery. It further signifies that the impacts are not dimensionless values. Beyond the ballpark loss estimate, economic impacts have a temporal dimension shaped by dynamic factors like accompanying hazards, such as earthquakes, damaged infrastructures, and affected population flux, often neglected due to data limitations. This study demonstrated that EC data could fill this data gap. However, gaps still exist, particularly in spatial and sectoral disaggregation of EC data. Hence, it would be helpful for researchers to have more frequently updated EC data with finer spatial resolution and sectoral classification to enable near-real-time impact monitoring and to have further studies investigating how monitoring these impacts could reduce losses and recovery time.

The spatial analysis from the NTLC maps can guide reconstruction planning. Reconstruction should avoid high-risk zones identified as affected areas in the NTLC

maps, which coincided with those devastated by past eruptions, such as in 1911 (Maso 1911). The NTLC map can also serve as a basis for assistance funds application, to which victims can apply based on the state of calamity declaration, often covering entire administrative units. Such broad coverage means that support is not always proportionate to the damage suffered. This study highlights the uneven impact distribution within a municipality, suggesting that the NTLC map can help direct aid to the most affected citizens.

6. Conclusion

This study analyzed the spatiotemporal distribution of economic impacts using daily EC and monthly NTL data. We computed ECL and NTLC as indicators of economic activity reduction. Results showed dynamic fluctuations and spatial heterogeneity in economic impacts. Temporal analysis showed that ECL had two loss peaks and two recovery valleys, congruent with power outages and restoration, road closures, earthquake occurrences, the number of affected populations, evacuees, and assistance distribution. This finding highlights the importance of temporal dimension analysis and frequent data updating to capture the dynamic impact developments. The spatial analysis showed significant NTL reductions at the intersection of ashfall extent and built-up areas. This analysis detected the municipalities with the most NTL reduction and mapped the most affected areas within municipalities. This finding can guide reconstruction planning and support prioritization.

We note that the results are limited to non-agricultural economic activities as these do not use substantial energy and emit significant light at night. Further, the results only cover the immediate impacts because the method cannot perform impact estimates for multiple hazards. This study also featured two separate analyses using EC for temporal distribution and NTL for spatial distribution due to the differences in scales. Future research could lengthen the study period to monitor long-term impacts, use finer resolution data with sectoral classification, and explore other ways to use EC in conjunction with NTL data.

To conclude, EC and NTL datasets can be alternative methods for analyzing the spatiotemporal distributions of immediate economic impacts from a volcanic eruption. This method features remote impact assessment for response and recovery planning, even with ongoing hazards. Using high-frequency EC and high-resolution NTL data, economic impact estimates of volcanic eruptions can be free from spatiotemporal blindness.

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Data availability

Data will be made available on request.

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